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Assignment-2 [CA-Software]

Scenario:- E-commerce platforms using micro-services and event Driven Architecture.

- Each microservice can be scaled independently allowing for more efficient resource allocation.
- They enable the use of different programming language, framework and databases for each service.
- If one microservice experiences issues, it won't bring down entire system.
- It allows for parallel development, reducing overall development time.
- EDA enable loose coupling between micro-services, allowing for greater flexibility and scalability.
- EDA provides a clear audit trail, making it easier to debug and troubleshoot issues.
- Example:- Payment Service, Inventory Service.

Application of SOLID principles —

- S Single Responsibility principle - One reason to change.
- O Open/Closed principle - open for extension, closed for modification.
- L Liskov Substitution principle - subtypes substitutable for base.
- I Interface Segregation principle - Clients shouldn't depend on unused interfaces.
- D Dependency Inversion - depend on abstractions not concretions.

Observing DRY and KISS —

• DRY

- central logging and authentication services reused across micro-services.
- shared validation libraries.

• KISS

- use of light weight message formats.
- Clear, simple service boundaries and API contracts.